Policy-as-Code Engine for Academic Compliance using Retrieval-Augmented Generation for UGC, AICTE, and College Bylaws

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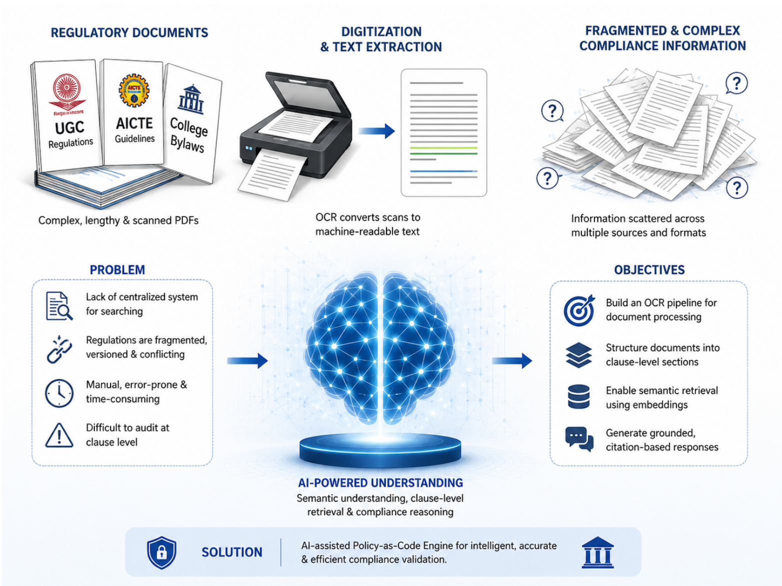
INTRODUCTION

Higher education institutions in India operate under complex and evolving regulations from UGC, AICTE, and internal bylaws. These regulations are often shared as lengthy or scanned PDFs, making manual interpretation and compliance checks difficult and error-prone. This project proposes a prototype AI-assisted compliance engine that combines: Retrieval-Augmented Generation (RAG) for intelligent document retrieval Policy-as-Code for automated rule-based reasoning A fully local open-source stack for privacy and cost efficiency



PROBLEM STATEMENT

The project addresses the lack of a centralized and reliable system for searching, interpreting, and verifying compliance across fragmented, versioned, and sometimes conflicting regulations issued by multiple authorities and distributed through heterogeneous PDF sources. Current compliance workflows are slow, manual, error-prone, and difficult to audit at the clause level.

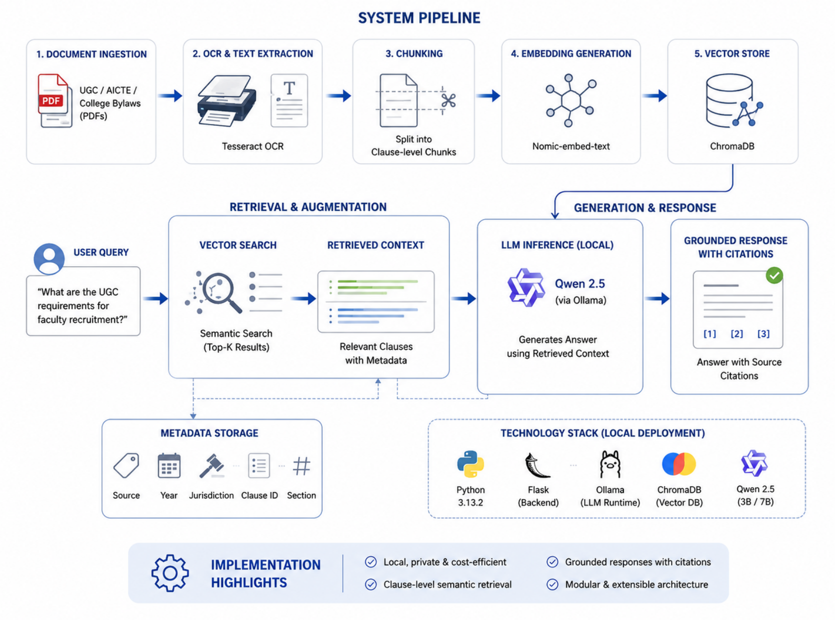
OBJECTIVES

Build an OCR pipeline to convert scanned and digital regulatory PDFs into structured, machine-readable text while preserving hierarchy and references. Split documents into clause-level sections with metadata like issuer, version, year, jurisdiction, and status. Implement semantic retrieval using nomic-embed-text, ChromaDB, and Qwen 2.5 via Ollama for grounded clause-based responses.

METHODOLOGY

The workflow begins by collecting UGC, AICTE, and institutional PDFs. Using Tesseract OCR and pdf2image, documents are converted into machine-readable text, cleaned, segmented into clauses, and enriched with metadata. Embeddings generated through nomic-embed-text are stored in ChromaDB.

When a user submits a query, the system retrieves relevant clauses through vector search, builds contextual input, and uses Qwen 2.5 to generate grounded responses with citations. The entire system runs locally for privacy and cost efficiency.



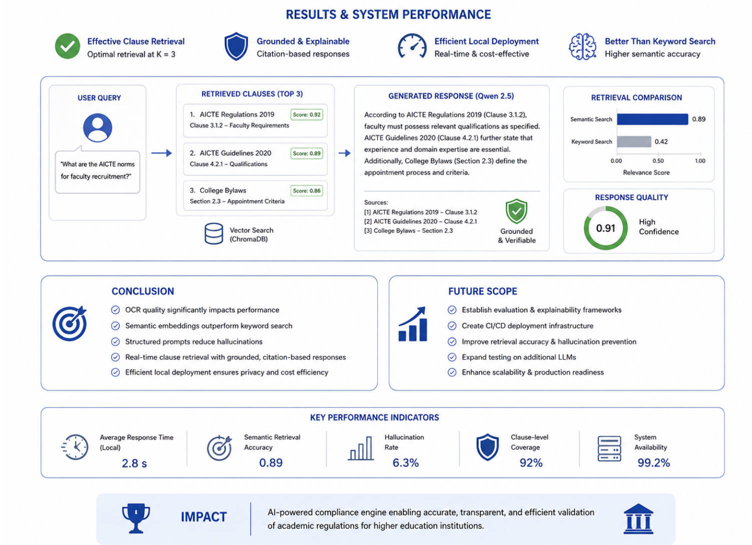
IMPLEMENTATION

The system uses Python 3.13.2 with a Flask backend and Ollama for local LLM inference, ensuring privacy and low operational cost. OCR and document processing are h`andled using Tesseract OCR, pdf2image, and Pillow for extracting text from scanned PDFs.

Qwen 2.5 (3B and 7B) models and nomic-embed-text are used for semantic retrieval and grounded response generation, while ChromaDB and SQLite manage vector search and metadata storage. The frontend is built with HTML, CSS, and JavaScript. The application runs on consumer hardware featuring an AMD Ryzen 7 4800H, NVIDIA RTX 3050 GPU (4GB VRAM), 16GB RAM, and Windows 11.

RESULTS

Testing showed that Qwen 3B performed better than Qwen 7B, with optimal retrieval at K = 3. Chain-of-thought prompting produced the best responses. The system achieved real-time clause retrieval, grounded citation-based responses, efficient local deployment, and better semantic retrieval than keyword search.



CONCLUSION AND FUTURE SCOPE

The project showed that OCR quality strongly impacts performance, semantic embeddings outperform keyword search, and structured prompting reduces hallucinations. Metadata-driven reasoning simplified compliance validation, while modular architecture and explainability improved system reliability. Future work includes enhancing evaluation frameworks, retrieval accuracy, hallucination prevention, CI/CD infrastructure, LLM testing, and production scalability.

REFERENCES

[1] A. Bernauer and R. A. Eisenberg, "Eiger: Auditable, executable, flexible legal regulations," 2022.

[2] H. Zhang et al., "Adaptable logical control for large language models," NeurIPS, 2024.

[3] C. Liang et al., "Neural symbolic machines," ACL, 2017.

[4] P. Lewis et al., "Retrieval-augmented generation for knowledge-intensive NLP tasks," NeurIPS, 2020.

[5–7] S. Geng et al., "Grammar-constrained decoding for structured NLP tasks,"